

PC1 (2023/2024)
ANALYSIS I
RESIST EXAM (2 hours)



- Mobile phones are strictly prohibited in the examination hall.
- Scientific calculators are not permitted during the exam.
- Only basic calculators, can be used.
- All answers must be well justified
- Answers written with a red pen or pencil will not be marked.
- Do not divide the page into two columns.

Exercise 1. (5 pts)

Let Ω be the subset of $\mathbb{R}$ defined by

$$\Omega = \left\{ \frac{m + n + 2mn}{m + n + mn + 1} : m, n \in \mathbb{N}_0 \right\},$$

where $\mathbb{N}_0 = \{0, 1, 2, \dots\}$.

Show that Ω is bounded and determine $\sup \Omega$, $\inf \Omega$, $\max \Omega$ and $\min \Omega$ when they exist.

Exercise 2. (5 pts)

Let for all $n \in \mathbb{N}$, α_n be the unique positive solution of the equation $x^n + x - 1 = 0$. Show that the sequence (α_n) is convergent and determine its limit.

Exercise 3. (5 pts)

Let $f, g : [0, 1] \rightarrow [0, 1]$ be two continuous functions such that $f \circ g = g \circ f$. Show that there exists $a \in [0, 1]$ such that $f(a) = g(a)$.

Exercise 4. (5 pts)

Let $f : [\frac{1}{2}, \frac{3}{2}] \rightarrow \mathbb{R}$ be the function defined by $f(x) = \frac{2x+1}{3x+1}$.

1. Calculate $f'(x)$ and $f''(x)$ for all $x \in [\frac{1}{2}, \frac{3}{2}]$.
2. Determine $f'([\frac{1}{2}, \frac{3}{2}])$ and $|f'|([\frac{1}{2}, \frac{3}{2}])$.
3. Determine the constant $k = \sup \{|f'(x)| : x \in [\frac{1}{2}, \frac{3}{2}]\}$.
4. Prove that for all $x, y \in [\frac{1}{2}, \frac{3}{2}]$ we have $|f(x) - f(y)| \leq k|x - y|$.
5. Solve the equation $f(x) = x$ in $[\frac{1}{2}, \frac{3}{2}]$.
6. By using questions 4 and 5 show that the sequence (v_n) defined by

$$v_0 = \frac{1}{2}, \quad v_{n+1} = f(v_n)$$

is convergent and determine its limit.